CTU CZECH TECHNICAL UNIVERSITY IN PRAGUE

BACHELOR‘S THESIS ASSIGNMENT

I. Personal and study details

Student's name: **Horák Jan** Personal ID number: **527481**

Faculty / Institute: **Faculty of Electrical Engineering**

Department / Institute: **Department of Computer Graphics and Interaction**

Study program: **Open Informatics**

Specialisation: **Computer Games and Graphics**

II. Bachelor’s thesis details

Bachelor’s thesis title in English:

Design and Implementation of a PHP Web Application Framework with AI Support for Website Generation and Management

Bachelor’s thesis title in Czech:

Návrh a implementace webového aplikačního frameworku v PHP s podporou umělé inteligence pro generování a správu webových stránek

Name and workplace of bachelor’s thesis supervisor:

Bc. Egor Ulianov Department of Computer Graphics and Interaction FEE

Name and workplace of second bachelor’s thesis supervisor or consultant:

Date of bachelor’s thesis assignment: **13.02.2026** Deadline for bachelor thesis submission: ____________

Assignment valid until: **19.09.2027**

Jiří Žára Digitálně podepsal(a) Jiří Žára Datum: 16.02.2026 13:54:45 ID: 30906

Head of department’s signature

Veronika Sobotíková Digitálně podepsal(a) Veronika Sobotíková Datum: 17.02.2026 07:07:12 ID: 31854

Vice-dean´s signature on behalf of the Dean

III. Assignment receipt

The student acknowledges that the bachelor’s thesis is an individual work.
The student must produce his thesis without the assistance of others, with the exception of provided consultations.
Within the bachelor’s thesis, the author must state the names of consultants and include a list of references.

17.02.2026

Date of assignment receipt

Horák Jan

Student’s signature

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Guidelines:

Building a website in a CMS typically requires a combination of developer work (structure, templates, integration) and content team work (copy, media). The goal of this thesis is to design and implement a prototype framework that can transform a natural-language request into a deployable website. Generative AI will act as an assistive layer in the system: it will propose structure, generate content, compose UI blocks, and prepare parts of templates or configuration - always with human control, review, and approval.
The thesis will focus on the design and implementation of a prototype framework in PHP that:
provides core building blocks of a modern web application (routing, controllers, middleware, templating, validation, persistence, authentication),
adds CMS capabilities on top of this core (website, content, media, and publishing management),
integrates a generative AI module that processes natural-language requests related to website look, structure, and content, and converts them into executable system changes.
The output will be a functional framework prototype and a demonstration application that practically validates the proposed solution. Within the MVP scope, the prototype will target generation and management of a multi-page presentation website of approximately 10-15 pages. The solution will use predefined page templates, specifically for a home page, a generic content page, an article list, an article detail, and a contact page. It will include full content management with CRUD operations, media management, and a publishing workflow with metadata and draft/published states. In terms of access control, the system must support at least the basic user roles: anonymous user, admin, and editor. The generated output must be responsive and include basic SEO metadata. At minimum, AI integration must cover three key scenarios: website structure design (especially menu and page types), generation of text and visual content for individual blocks, and proposing modifications of an existing page based on a natural-language request.
The student will demonstrably validate the prototype on at least three demonstration websites of different types (e.g., personal, corporate, content-driven), each with at least 8-12 pages. For each website, the full workflow will be documented - from initial natural-language input through structure design and content generation to publication and subsequent edits.
The validation will include:
(1) functional testing of key framework and CMS components (routing, roles, CRUD, publishing),
(2) testing of AI scenarios (structure design, text generation, existing-page modifications),
(3) evaluation using simple metrics (requirement coverage, number of manual interventions, time to publishable draft, responsiveness, and basic SEO), and
(4) short user testing with at least 5 evaluators.

The final output will include a concise evaluation report summarizing results and limitations of the solution.

Bibliography / sources:

MINAEE, Shervin; MIKOLOV, Tomáš; NIKZAD, Narjes; CHENAGHLU, Meysam; SOCHER, Richard; AMATRIAIN, Xavier; GAO, Jianfeng. Large Language Models: A Survey. arXiv preprint, 2024. DOI: 10.48550/arXiv.2402.06196

ANJUM, Nirjhor; ALAM, Shamimul. A Comparative Analysis on Widely Used Web Frameworks to Choose the Requirement Based Development Technology. International Advanced Research Journal in Science, Engineering and Technology (IARJSET), 2019, 6th year, n. 9, p. 16–24. DOI: 10.17148/IARJSET.2019.6902.

PHP DOCUMENTATION GROUP. PHP: Documentation (PHP Manual). php.net [online]. [cit. 2026-02-11]. Available: php.net/docs.php